Tavishi Session 22

Sunday, July 12, 2026 8:43 PM

Basics
- Variables
- Data Types
- Type Conversion
- Input and Output
- Comments
- Operators

Control Flow
- if
- if-else
- if-elif-else
- Nested Conditions
- Match Case (optional)

Loops
- for Loop
- while Loop
- Nested Loops
- Loop Control Statements (break, continue, pass)

Strings
- String Indexing
- String Slicing
- String Methods
- String Formatting

Collections
- Lists
- Tuples
- Sets
- Dictionaries

Collection Operations
- List Methods
- Tuple Operations
- Set Methods
- Dictionary Methods

Functions
- Function Definition
- Parameters and Arguments
- Return Statement
- Default Arguments
- Keyword Arguments
- Variable Length Arguments (*args, **kwargs)
- Scope (Local and Global)
- Lambda Functions
- map()
- filter()
- reduce()

String formatting :-

name = "Tavish"
print ("Hello", name)

O/P — Hello Tavishi

name = "Tavishi" print ("name")

```
name="Tavishi"
print("Hello",name)

# String formatting
_name="Azaan"
age=24
profession="Software Engineer"
```

```
working="Remotely"

print(_name,"is a tutor at Codeyoung whose age is",age,"he is working",working,"as a", profession)

#String formatting using f string
print(f"{_name} is a tutor at Codeyoung whose age is {age} he is working {working} as a {profession}")
```

Data Types

Imagine a bank account.

A bank account contains different kinds of information:

- Account Holder Name → `"Shanmuka"`
- Account Balance → `50000.75`
- Account Active → `True`
- Transaction IDs → `[101, 102, 103]` → Collection

All these values are different.

Python needs to know:

- Which value is text?
- Which value is a number?
- Which value is a yes/no answer?
- Which value is a collection?

This is where **Data Types** become important.

Variables ↑

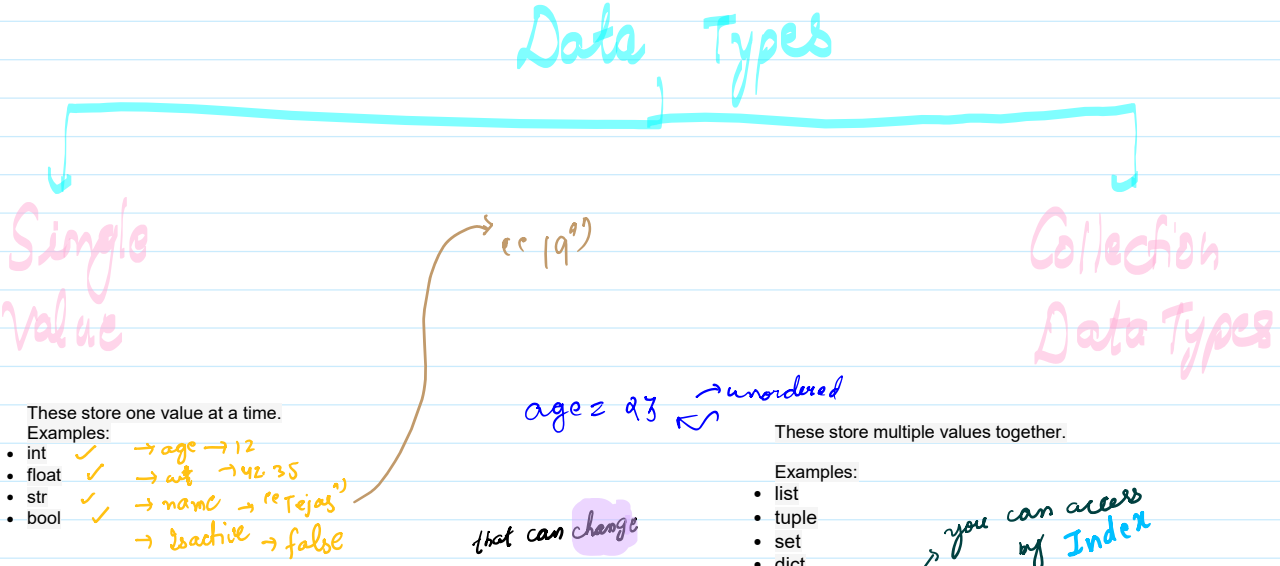
What kind of data it stores →

Variables:-

name = "Sophie"

age = 16

wt = 45.75



Data Type	Example	Mutable/Immutable	Ordered/Unordered	Real-Life Example
int	100	Immutable	N/A → Not Applicate	Roll Number
float	99.5	Immutable	N/A	Product Price
str	"Python"	Immutable	Ordered	Student Name
bool	True	Immutable	N/A	Login Status
list	[1,2,3]	Mutable	Ordered	Shopping Cart
tuple	(1,2,3)	Immutable	Ordered	Coordinates
set	{1,2,3}	Mutable	Unordered	Unique IDs
dict	{"name":"Rahul"}	Mutable	Insertion-Ordered	Student Record

single

collection

List ← collection data types

- List is a group of similar or different data types.

Ex- marks = [14, 15, 8, 9, 11]

random_hobbies_age = ["Volleyball", 24, TRUE]

"Volleyball" → str
24 → int
TRUE → boolean
I know Python

nested list = [[1, 2, 3]]

loops
loop → inside → loop

A List allows us to store multiple values in a single variable.

Lists are:
✓ Ordered → we can access it using index
✓ Mutable → It can be change
✓ Allow Duplicate Values

```
subjects = ["Math", "Science", "English"]

# Print the entire list of subjects
print(subjects)
# Print the data type of the 'subjects' variable
print(type(subjects))
```

Accessing a list

```
#~ Accessing a List using index
subjects = ["Math", "Science", "English"]
#           0       1         2
print(subjects[0])
print(subjects[1])
print(subjects[2])
```

Modifying a List

```
subjects = ["Math", "Science", "English"]
#           0       1         2


subjects[1]="Physics"
print(subjects)
```